Kuan-Fu Feng

contact e-mail: kuanfu.feng@utah.edu

GitHub [click] | Linkedin Profile [click] | Google Scholar [click]

Research Seismologist with 10+ years of specialized experience in time series analysis, signal processing, and monitoring. Proficient in scientific computing, programming, cloud computing, and open-source tools. Capable of translating models into practical workflows and dedicated to applying research findings to engineering solutions.

Skill

Software Engineering	Linux/Unix, Git/GitHub, Jupyter Notebook, Visual Studio Code
Programming Languages	Python, Fortran, C, C++, Bash Script
Scientific/Technical Skills	Time series analysis, Signal processing, Inversion methods, Statistical analysis
Data visualization	ParaView, Matplotlib, Seaborn, Adobe Illustrator
Cloud/HPC Environment	AWS (S3, EC2, Jupyter), parallel scripting, workflow automation

Applied Research Impact

- Designed scalable seismic processing pipelines for large observational datasets.
- Developed monitoring frameworks capable of detecting hydrologically/tectonically induced subsurface velocity changes.
- Implemented physical-based and statistical methods to examine environmental effects on multiscale (crust/critical zones).
- Contributed to research relevant to groundwater–surface interaction and subsurface permeability evolution.

Professional Experience

Postdoctoral Research Associate

University of Utah, USA — *Dec 2024 - present*
University of Washington, USA — *Jun 2023 - Nov 2024*
University of Utah, USA — *Jun 2022 - May 2023*

- Led time-lapse seismic monitoring studies integrating multi-scale geophysical and environmental datasets to characterize subsurface fluid-driven property changes.
- Processed 10+ TB of ambient noise data using automated HPC/cloud workflows for scalable subsurface investigations.
- Developed data-driven and inversion-based models to quantify hydrologically induced seismic velocity variations.
- Contributed to open-source software development and delivered workshops on cloud-enabled scientific computing (AWS).

Graduate Research Assistant

National Taiwan University, Taiwan — *Sep 2017 - Jan 2022*
National Taiwan University, Taiwan — *Sep 2014 - Jun 2016*

- Designed and implemented ambient noise–based seismic monitoring systems to track tectonic and hydrological processes.
- Performed time-frequency analysis of multi-year datasets to resolve crustal velocity changes and fluid migration signals.
- Supported seismic array deployment and analyzed earthquake source processes and structural models.

Research Assistant

Institute of Earth Sciences, Academia Sinica, Taiwan — *Oct 2016 - Aug 2017*

- Maintained and tested Real-time Earthquake Moment Tensor Monitoring System.
- Constructed finite-fault models of subduction zone events using broadband waveform modeling.

Education

PhD	*Geosciences, National Taiwan University, Taiwan*	*Sep 2017 - Jan 2022*
Master	*Geosciences, National Taiwan University, Taiwan*	*Sep 2014 - Jun 2016*
Bachelor	*Earth and Environmental Sciences, National Chung Cheng University, Taiwan*	*Sep 2010 - Jan 2014*

Service & Leadership

Peer-reviewer

Reviewed manuscripts for *Geophysical Research Letters* (2), *Journal of Geophysical Research* (5), *Nature Communications* (1), *Earth, Planets and Space* (1), *Tectonophysics* (1).

Workshop instructor

Earth and Space Sciences, University of Washington — *2024*

- Delivered workshop on Ambient Noise Seismology in the Cloud [SCOPED Workshop (link)].
- Instructed Data Mining on the Cloud 101 at the Seismological Society of America Meeting [SSA Tutorials (link)].

Lecture Convenor and Instructor for Undergraduate Summer Interns

Institute of Earth Sciences, Academia Sinica, Taiwan — *Summer 2018 and 2019*

- Managed logistics for an institute's summer lecture series aimed at undergraduate students.
- Designed and delivered instruction on earthquake seismology and time-series analysis

Peer-reviewed Publication

Cloud Seismology

[C.2] Ni, Y., et al. (2025) **A Review of Cloud Computing and Storage in Seismology.** *Geophysical Journal International*, ggaf322. [https://doi.org/10.1093/gji/ggaf322]

[C.1] Denolle, M., et al. (2025) **Training the Next Generation of Seismologists: Delivering Research-Grade Software Education for Cloud and HPC Computing through Diverse Training Modalities.** *Seismological Research Letters*. [https://doi.org/10.1785/0220240413]

Ambient Noise and Monitoring

[A.4] Kidiwela, M., et al. (2026, accepted) **Active Protothrusts and Fluid Highways: Seismic Noise Reveals Hidden Subduction Dynamics in Cascadia** *Science Advances* [in press, https://arxiv.org/abs/2505.15731]

[A.3] Feng, K.-F., et al. (2026) **A decadal survey of the near-surface seismic velocity response to hydrological variations in Utah, United States.** *Journal of Geophysical Research: Solid Earth.*, 131(1), e2024JB030689.[https://doi.org/10.1029/2024JB030689]

[A.2] Feng, K.-F., et al. (2021) **Controls on Seasonal Variations of Crustal Seismic Velocity in Taiwan Using Single-station Cross-component Analysis of Ambient Noise Interferometry.** *Journal of Geophysical Research: Solid Earth*, 126(11), e2021JB022650. [https://doi.org/10.1029/2021JB022650]

[A.1] Feng, K.-F., et al. (2020) **Detecting Pre-eruptive Magmatic Processes of the 2018 Eruption at Kilauea, Hawaii Volcano with Ambient Noise Interferometry.** *Earth, Planets and Space*, 72, 74. [https://doi.org/10.1186/s40623-020-01199-x]

Earthquake Source and Seismogenic Structure

[S.3] Hsu, Y.-F., et al. (2020) **Evidence for Fluid Migration During the 2016 Meinong Taiwan Aftershock Sequence.** *Journal of Geophysical Research: Solid Earth*, 125(9), e2020JB019994. [https://doi.org/10.1029/2020JB019994]

[S.2] Lee, S.-J., et al. (2018) **Composite Megathrust Rupture from Deep Interplate to Trench of the 2016 Solomon Islands Earthquake.** *Geophysical Research Letters*, 45(2), 674-681. [https://doi.org/10.1002/2017GL076347]

[S.1] Brown, D., et al. (2015) **Imaging High-pressure Rock Exhumation in Eastern Taiwan.** *Geology*, 43(7), 651-654. [https://doi.org/10.1130/G36810.1]